

ALI RAZA KHAN

AI/ML Engineer — Data Science · Agentic AI · Financial ML
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PROFESSIONAL SUMMARY

AI/ML engineer building end-to-end systems that take reliable, auditable actions, from regulated risk scoring to agentic explainability. Hands-on with classical ML (XGBoost, Logistic Regression), transformer fine-tuning, mathematical optimization, and LLM/RAG agents grounded in deterministic model outputs rather than free-text generation. Production C++/real-time background from game engineering brings performance discipline and clean, testable systems to applied AI. Comfortable owning ambiguous, high-impact problems end to end and writing things down.

TECHNICAL SKILLS

Languages: Python, C++, C#, SQL, JavaScript

AI/ML: XGBoost, LightGBM, Scikit-learn, PyTorch, Hugging Face Transformers, DistilBERT/ViT fine-tuning, SHAP explainability, OpenCV, NumPy, Pandas

Agentic AI: LangGraph, RAG, FAISS, Ollama (llama3.2, nomic-embed-text), retrieval-grounded reasoning, counterfactual generation

Optimization: IBM CPLEX (docplex), constrained portfolio allocation

Backend & Deploy: FastAPI, REST APIs, NodeJS, React (Vite), model serialization/versioning, prediction logging, Docker, Git

AI / ML & DATA SCIENCE PROJECTS

Credit Decisioning System with Agentic Explainability | [Repo](#)

- **Built** an end-to-end credit scoring pipeline on 307,511 applications (122 raw → 84 engineered features, 8.07% default rate); benchmarked a Logistic Regression challenger (AUC 0.748, Gini 0.495, KS 0.367) against an XGBoost champion (AUC 0.769, Gini 0.538, KS 0.404), handling class imbalance with scale_pos_weight.
- **Added** SHAP explainability, global beeswarm/bar summaries and per-applicant waterfalls, surfacing EXT_SOURCE_3, EXT_SOURCE_2, and loan term as dominant drivers.
- **Formulated** a portfolio-allocation optimization in IBM CPLEX (docplex): selected 20 of 500 approved loans against a \$10M capital budget at full utilization, yielding 22.4% return on capital at 9.7% average PD.
- **Engineered** a 3-node LangGraph RAG agent (retrieve applicant → retrieve policy → generate answer) over credit-policy docs using Ollama + FAISS, producing grounded, actionable counterfactuals for loan-officer queries.

Walmart Demand Forecasting Pipeline | [Repo](#)

- **Built** an end-to-end forecasting pipeline on 421,570 rows of weekly sales; merged three sources and engineered lag (1/4/52-week), rolling-mean, and calendar features across 45 stores and 81 departments.
- **Benchmarked** XGBoost vs LightGBM with early stopping on a chronological split (MAE \$1,374, RMSE \$3,028); Sales_Lag_52 emerged as the dominant signal.

Transformer & Vision Projects | DistilBERT · ViT | [Repo](#)

- **Fine-tuned** DistilBERT (AG News multi-class classification) and a Vision Transformer (AI-vs-real image detection); deployed both behind FastAPI backends with React frontends for live inference.

PROFESSIONAL EXPERIENCE

Software Engineer - I | Devsinc

Jun 2025 – July 2026

- **Design** and implement real-time C++/Blueprint gameplay and AI behavior pipelines within a professional product team; apply component-based architecture, code review, and version-control discipline.

Game Development Intern | Dreamcast Interactive · Mindstorm Studios

2024 – 2025

- **Built** gameplay prototypes and shipped features in C++/Blueprints and Unity (C#), including multiplayer networking (replication, RPCs) and custom UI systems.

EDUCATION & CERTIFICATIONS

BS Computer Science | Lahore Garrison University

2021 – 2025

Certifications: NVIDIA Omniverse Extensions Development · Rookie Game Jam 2024 · Web Development (LUMS)